PhaseAtlas
From PDE animations to reproducible evidence

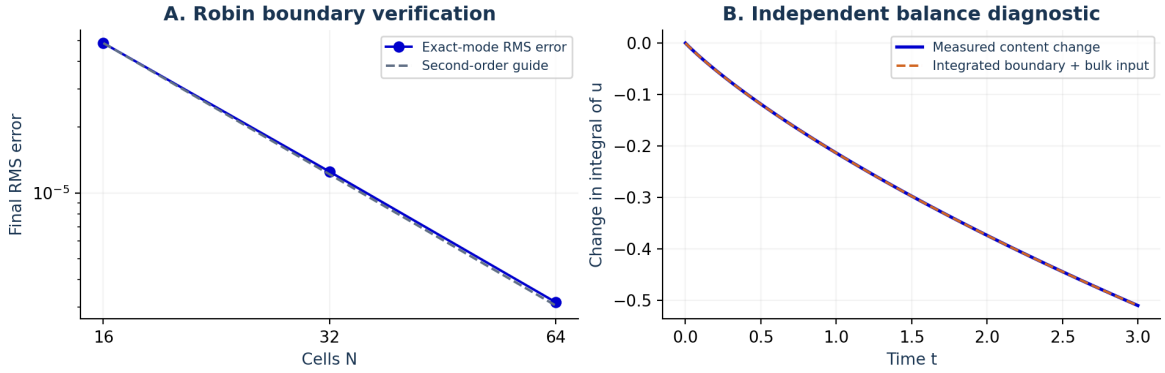
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18 October 2026 | Pre-event project snapshot

The problem

An attractive PDE animation can conceal a wrong boundary condition, inadequate resolution or a misleading interpretation of a travelling wave. Researchers need to inspect the calculation behind a figure and give someone else enough information to reproduce it.

The contribution

PhaseAtlas is a browser-based scientific workbench that connects editable PDE models to animated experiments, numerical checks and portable evidence. The workflow combines three-level space/time studies, travelling-wave motion and deformation diagnostics, and parameter maps whose cells retain their trajectories. Supported second-order bounded models include Dirichlet, Neumann and passive Robin conditions; 1D fourth-order models accept two configurable linear boundary conditions at each end.



Executed benchmarks: Robin exact-mode errors decrease with refinement (left); measured thermal-content change tracks integrated model input for a cooling rod (right). The latter also depends on saved-frame quadrature.

Demonstrated result and practical value

For $u_t = 0.1\, u_{xx}$ on [-1,1], the Robin eigenmode test at t = 0.2 gives RMS errors of 4.863e-5, 1.250e-5 and 3.148e-6 on 16, 32 and 64 cells; the three-level observed order is 1.99995. All 90 current mathematical and numerical tests pass. A dimensionless cooling example relates wall exchange and diffusivity to mean temperature, while the MAE03 fourth-order front connects the workflow to nonlinear-wave research. These examples help users design comparisons, identify numerical artifacts and exchange complete experiments.

What another researcher receives

Equation and boundary editors; 1D/2D visualization for supported models; animation replay; refinement and boundary diagnostics; finite-time parameter scans; JSON/CSV/figure exports; and an independent Python reference. Computation runs locally in browser workers. No account or runtime LLM API key is required.

Scope and provenance

This is independently implemented, AI-assisted scientific software. It does not solve arbitrary PDEs or establish a new stability theorem. The cooling example is a model study, not an experimentally calibrated engineering prediction. The earlier MAE03 research and this 18 October software snapshot are pre-event work; any October 21-22 contribution must be documented separately.

Project: https://github.com/NIU-55383/PhaseAtlas
Companion: full technical article, data, scripts and numerical methods in the submission package.